PROJECT 6: DEPLOY A WEB SERVER (APACHE) ON AN EC2 INSTANCE REPORT

INTRODUCTION

Amazon Elastic Compute Cloud (Amazon EC2) is a core cloud computing service provided by Amazon Web Services (AWS) that enables users to launch and manage virtual servers in the cloud. EC2 allows organizations and developers to deploy applications, host websites, and run workloads without maintaining physical infrastructure.This project focuses on launching an EC2 instance, configuring network access through security groups, installing the Apache HTTP Server, and hosting a custom webpage. The project demonstrates the process of deploying a web server on AWS and making it accessible through a public IP address.

OBJECTIVES

The primary objectives of this project are:

- Launch an Amazon EC2 instance
- Configure security group rules for web access
- Install the Apache HTTP Server
- Start and enable the Apache service
- Deploy a custom webpage
- Access the website using a public IP address
- Understand the fundamentals of web server deployment on AWS

OVERVIEW OF AMAZON EC2 AND APACHE WEB SERVER

Amazon EC2

Amazon Elastic Compute Cloud (EC2) is a web service that provides scalable virtual servers in the cloud. EC2 allows users to quickly provision computing resources and run applications on demand.

Security Groups

Security Groups act as virtual firewalls that control inbound and outbound traffic for EC2 instances. They help secure cloud resources by allowing only authorized network communication.

Apache HTTP Server

Apache HTTP Server, commonly known as Apache, is one of the most widely used open-source web servers. It enables websites and web applications to be hosted and accessed over the internet.

HTTP Protocol

Hypertext Transfer Protocol (HTTP) is the communication protocol used to transfer web pages between web servers and client browsers. By default, HTTP uses Port 80.

Public IP Address

A Public IP Address allows users to access services hosted on an EC2 instance from the internet.

METHODOLOGY

Step 1: Launching the EC2 Instance

An EC2 instance named Apache-WebServer was launched using the AWS Management Console. The Amazon Linux 2023 Amazon Machine Image (AMI) was selected along with the t3.micro instance type. A key pair was configured to enable secure access to the instance.

The instance was successfully launched and verified to be in the running state.

Step 2: Configuring Security Group Rules

A security group was configured to control network access to the EC2 instance.

The following inbound rules were configured:

- SSH (Port 22) for remote administration
- HTTP (Port 80) for web traffic

These rules allowed administrators to manage the server and enabled users to access the hosted webpage through a browser.

Step 3: Installing the Apache Web Server

The EC2 instance was accessed using EC2 Instance Connect.

System packages were updated using:

- sudo dnf update -y

The Apache HTTP Server package was installed using:

- sudo dnf install httpd -y

The installation completed successfully and Apache was added to the server environment.

Step 4: Starting and Enabling the Apache Service

The Apache service was started using:

- ✧ sudo systemctl start httpd

The service was configured to start automatically during system boot using:

- ✧ sudo systemctl enable httpd

Service status verification confirmed that Apache was active and running successfully.

Step 5: Deploying and Testing the Webpage

A custom HTML webpage containing project information was created and placed in the default Apache web directory.

The website was accessed through the EC2 public IP address using a web browser.

Successful webpage loading confirmed that:

- ✧ Apache was functioning correctly
- ✧ HTTP traffic was allowed through the security group
- ✧ The EC2 instance was serving web content successfully

RESULTS

The following outcomes were successfully achieved:

- ✧ Launched an Amazon EC2 instance
- ✧ Configured security group rules for SSH and HTTP
- ✧ Installed the Apache HTTP Server
- ✧ Started and enabled the Apache service
- ✧ Hosted a custom webpage on the server
- ✧ Successfully accessed the webpage through the public IP address
- ✧ Verified web server functionality and network connectivity

KEY LEARNINGS

Throughout this project, several important cloud computing concepts were learned:

- ✧ Amazon EC2 provides scalable cloud-based virtual servers
- ✧ Security Groups control network access to EC2 instances
- ✧ Apache HTTP Server can be installed and managed on Linux servers
- ✧ HTTP communication occurs through Port 80
- ✧ Linux services can be managed using systemctl commands
- ✧ Public IP addresses allow hosted applications to be accessed from the internet
- ✧ AWS simplifies web server deployment and infrastructure management

ADVANTAGES

- ✧ Amazon EC2 and Apache provide several benefits:
- ✧ Rapid server provisioning and deployment
- ✧ Cost-effective cloud infrastructure
- ✧ Scalability and flexibility
- ✧ Easy web hosting and application deployment
- ✧ Secure network access through security groups
- ✧ High availability and reliability
- ✧ Integration with other AWS services

LIMITATIONS

Despite its advantages, the deployment also has certain limitations:

- ✧ HTTP traffic is not encrypted
- ✧ Public IP addresses may change after instance restart
- ✧ Misconfigured security groups can expose resources
- ✧ Continuous monitoring is required for production environments
- ✧ Additional configuration is required for HTTPS implementation
- ✧ Server maintenance and updates remain necessary

CONCLUSION

This project provided practical hands-on experience in deploying a web server using Apache on an Amazon EC2 instance. The implementation involved launching cloud infrastructure, configuring security groups, installing and managing Apache services, and hosting a custom webpage accessible through a public IP address.

The successful completion of this project demonstrated how AWS EC2 can be used to host web applications and websites in a cloud environment. It also reinforced important cloud computing concepts such as infrastructure provisioning, network security, service management, and web hosting.

Overall, the project strengthened practical AWS skills and provided valuable experience in deploying and managing cloud-based web servers.